MIL-T-51023D
29 December 1967
SUPERSEDING
MIL-T-51023C(MU)
29 September 1966

MILITARY SPECIFICATION

TABLETS, COMPOUND 34

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers tablets containing Compound 34, p-amino-o-ethoxy-6-sulfo-diphenylamine.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

JJJ-S-791 - Sugar, Refined and Brown; Beet or Cane.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

B5-77-168 - Vial, Green, Assembly and Detail.
DL136-13-500 - Tester, Friability Q33

FSC 6665

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

THE UNITED STATES PHARMACOPOEIAL CONVENTION, INC.

Pharmacopeia of the United States, XVI Revision

(Copies of the US Pharmacopeia may be obtained from the Mack Publishing Company, Easton, Pennsylvania 18042.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standard

D1193-56 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Tablet composition. Each tablet shall be double convex, 7/32 ± 1/32 inch in diameter, (see 6.4) and shall consist of:

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Material	Description	Milligrams
Compound 34*	Recrystallized, melting at 250° to 255°C and indicating no absorbance from 460 to 700 mμ when tested as a 0.001M methanol solution against a methanol blank	2.5 ± 0.2
Magnesium stearate	Technical grade	0.5
Sugar	JJJ-S-791, type 1(b)(2)	remainder

*When Compound 34 is furnished by the Government, the supplier shall reprocess the material to meet the above requirements.

3.2 Assembly. The tablets shall be assembled into vials in accordance with Drawing B5-77-168.

3.3 Compound 34 content. After aging for two weeks at 65° ± 2°C, the tablet shall contain 2.5 ± 0.2 milligram (mg) of Compound 34 when tested in accordance with 4.2.4.2.

3.4 Disintegration time. Before and after aging for two weeks at 65° ± 2°C, the tablet shall dissolve in no more than 10 minutes when tested in accordance with 4.2.4.3.

3.5 Friability. Before and after aging for two weeks at 65° ± 2°C, the tablet shall lose no more than 5 percent in weight when tested in accordance with 4.2.4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Regular production.

4.2.1 Lotting. A lot shall consist of the vials of Compound 34 produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions and containing no more than one batch of Compound 34 (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Vials of Compound 34 shall be taken at random from each lot for testing in accordance with table 1.

4.2.3 Inspection.

4.2.3.1 For examination. Sample vials of Compound 34 shall be inspected in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For tests. One tablet from each vial shall be used for each test in accordance with 4.2.4 and table 1.

Table 1. Sampling and testing

Test	Sampling (MIL-STD-105)		Test Paragraph
	Inspection Level	AQL Percent defective	
Size	S4	1.0	4.2.4.1
Compound 34	S2	2.5	4.2.4.2
Disintegration time			4.2.4.3
Before aging	S4	1.0	
After aging	S4	1.0	
Friability			4.2.4.4
Before aging	S4	1.0	
After aging	S4	1.0	

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4.2.3.3 Classification of defects.

(a) Vial, green, assembly and detail (Dwg B5-77-168).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Vial cracked or broken
102	Tablets broken or powdered
103	Cork damaged or fit incorrect
104	Coating missing or color incorrect
105	Wax dip missing or inadequate
106	Internal contamination
107	Component missing
Minor:	AQL 2.5 percent defective
201	Vial length and diameter incorrect
202	Cotton missing or tablets loosely packed
203	External contamination
204	Marking missing or illegible

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Form and size. Select tablets in accordance with 4.2.2.2 and inspect for compliance with 3.1 for form and size. After this test, return tablets to original vials from which obtained.

4.2.4.2 Compound 34 content after aging.

(a) Standard concentration curve. The Compound 34 used in the preparation of the standard concentration curve shall have a melting point between 250° to 255°C and when tested as a 0.001M methanol solution against a methanol blank shall indicate no absorbance in the 460 to 700 mμ range. The standard concentration curve shall be prepared in the following manner:

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(1) Accurately weigh approximately 35 mg of Compound 34, dissolve it in approximately 70 milliliters (ml) of water in a 100 ml volumetric flask by shaking at room temperature over 1 hour. Dilute to exactly 100 ml with water and mix well. Transfer measured volumes of this standard solution, equivalent to approximately 1.5, 2.0, 2.5, 3.0, and 3.5 mg of Compound 34, to five 100 ml volumetric flasks. Add water to each, to bring the volume to 20 ml. Then treat each standard as follows:

(2) Adjust the acidity to pH 3 or less with hydrochloric acid. (Two drops of the concentrated acid has been found sufficient.) Cool to below 10°C in an ice bath and add dropwise a 1 percent solution of sodium nitrite in water until a blue color is obtained on starch-iodide paper. (Four to five drops have been found sufficient.) Add rapidly 5 ml of a solution (previously cooled to below 10°C) containing 0.05 gram of resorcinol and 0.25 gram of sodium hydroxide. Swirl and let the mixture stand for 5 minutes in the ice bath. Dilute to exactly 100 ml with water and mix well.

(3) Without delay dilute 5 to 50 with water and measure the absorbence of this solution at 465 mμ in a 1 centimeter cell versus water set to zero absorbence.

(4) Plot absorbence versus mg Compound 34 on rectangular coordinate graph paper. A straight line shall result. This is the standard curve.

(b) Procedure. Take one tablet and dissolve it in 20 ml of water in a 100 ml volumetric flask. Then follow the procedure for the preparation of standards beginning with 4.2.4.2(a)(2). From the standard curve, read the weight of Compound 34 in each tablet.

4.2.4.3 Disintegration before and after aging. Test each tablet with the disintegration apparatus and procedure described in the US Pharmacopeia, sixteenth revision, page 934. The acceptance/rejection criteria of this specification shall apply in lieu of the criteria specified in the US Pharmacopeia.

4.2.4.4 Friability before and after aging. Brush the sample tablets with a camel's hair brush to remove all adherent powder, and weigh each tablet to the nearest 0.1 mg. Place the tablets in tubes mounted on the drive wheel of Tester, Friability Q33 (DL 136-13-500) and with the axis of the wheel horizontal, rotate at a speed not exceeding 4 ± 1/4 rotations per minute (rpm) until the tablets have sustained 500 drops. Remove the tablets from the tube, brush thoroughly, and weigh each tablet to the nearest 0.1 mg.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The vials of Compound 34 tablets shall be packaged to provide adequate protection against breakage or other damage in shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Vials of Compound 34, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Identification marking. Each package shall be marked with the following:

Tablets, Compound 34
Lot number
Number of vials in package
Expiration date (3 years from date of manufacture)
Manufacturer
Date of manufacture.

6. NOTES

6.1 Intended use. The tablets covered by this specification are intended for use in detector kits for the detection of G-agents.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 Weight of Tablet. The weight of each tablet should be approximately 105 ± 7 milligrams.

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6.5 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation: E29).

6.6 International interest. Certain provisions of this specification are the subject of international standardization agreement ABCA-ARMY-STD. 94. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodians:

Army - MU
Navy - SH
Air Force - 82

Preparing activity:

Army - MU (EA)

Reviewer activities:

Army - MU
Navy - SH
Air Force - 82

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User activities:

Army - MC
Navy - OS, SH
Air Force - 82

International interest (see section 6)

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